

C3D-GAN: Conditional 3D GAN for Echocardiogram Video Generation

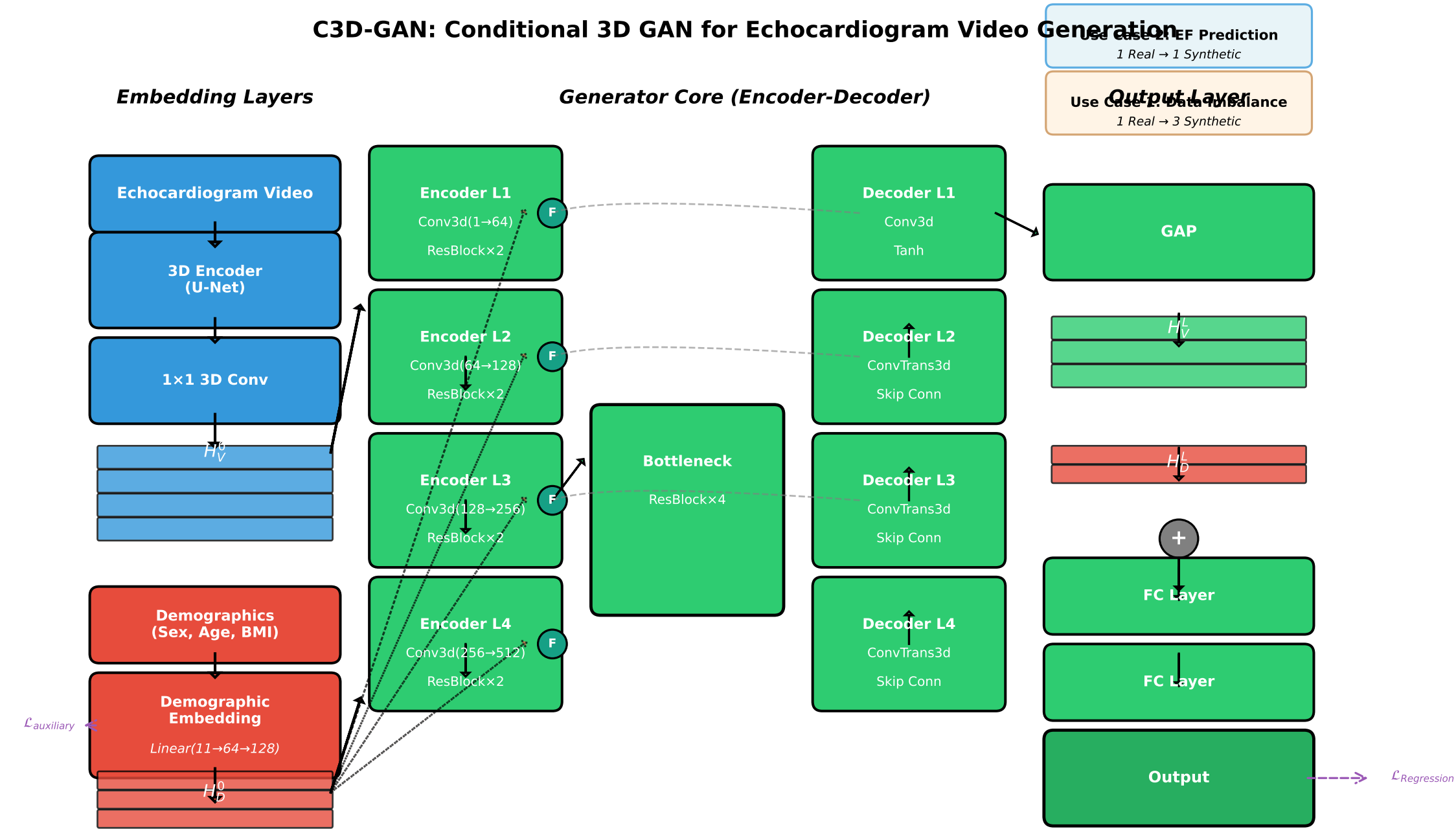


Fig. 1: C3D-GAN consists of: (1) the video embedding layers from echocardiograms, (2) the demographic embedding layers, (3) the generator core with encoder-decoder layers and demographic fusion, and (4) the output layers. The embedding layers extract latent variables of each modality, and the generator core captures their interactions through multi-level demographic fusion. Two use cases: (a) 1 real → 3 synthetic videos for addressing data imbalance, (b) 1 real → 1 synthetic video for EF prediction.